

MC UNIT-1 (Important Questions)

1. Find the rank of the given Matrix

(i)
$$\begin{bmatrix} 0 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \end{bmatrix} \quad (X)$$

Solⁿ finding rank of the matrix using NORMAL FORM

$$C_1 \rightarrow C_1 + C_2 \quad \begin{bmatrix} 1 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 3 & 1 & 3 & 1 \end{bmatrix}$$

$$\begin{aligned} R_2 &\rightarrow R_2 - 4R_1 \\ R_3 &\rightarrow R_3 - 3R_1 \end{aligned} \quad \begin{bmatrix} 1 & 1 & 2 & -2 \\ 0 & -4 & -6 & 14 \\ 0 & -2 & -3 & 7 \end{bmatrix}$$

$$R_3 \rightarrow 2R_3 - R_2 \quad \begin{bmatrix} 1 & 1 & 2 & -2 \\ 0 & -4 & -6 & 14 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} C_2 &\rightarrow C_2 - C_1 \\ C_3 &\rightarrow C_3 - 2C_1 \\ C_4 &\rightarrow C_4 + 2C_1 \end{aligned} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -4 & -6 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} C_3 &\rightarrow 4C_3 - 6C_2 \\ C_4 &\rightarrow 2C_4 + C_2 \end{aligned} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$C_2 \div -4 \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = [I_2 \ 0]$$

$$\therefore \rho(A) = 2.$$

(order of Identity matrix = rank of the matrix)

(ii) rank of the matrix using Normal form

$$\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

Solⁿ finding rank using Normal form

$$R_1 \leftrightarrow R_2 \quad \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

$$\begin{aligned} R_3 &\rightarrow R_3 - 3R_1 \\ R_4 &\rightarrow R_4 - R_1 \end{aligned} \quad \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \end{bmatrix}$$

$$\begin{aligned} R_3 &\rightarrow R_3 - R_2 \\ R_4 &\rightarrow R_4 - R_2 \end{aligned} \quad \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} C_3 &\rightarrow C_3 - C_1 \\ C_4 &\rightarrow C_4 - C_1 \end{aligned} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} C_3 &\rightarrow C_3 + 3C_2 \\ C_4 &\rightarrow C_4 + C_2 \end{aligned} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = [I_2 \ 0]$$

$$\therefore \rho(A) = 2$$

Q find Inverse of the matrix using elementary row operations.

$$A = \begin{bmatrix} 1 & -2 & -3 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Solⁿ finding Inverse of the given matrix using GAUSS JORDAN METHOD.

Take $A = IA$

$$\begin{bmatrix} 1 & -2 & -3 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

Applying elementary row operations on LHS matrix A to reduce the matrix A to Identity matrix)

$$R_1 \rightarrow R_1 + R_2 \quad \begin{bmatrix} 1 & 0 & -3 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$R_1 \rightarrow R_1 + R_3 \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$R_2/2 \quad R_3/3 \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix} A$$

$$I = B A$$

as $A \cdot A^{-1} = I$

here $A^{-1} = B$

$$\therefore A^{-1} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix}$$

IV. approximation

③ Solve up to three decimal using gauss seidel iteration.

$$\begin{aligned} \text{(i)} \quad & 10x + y + z = 12 \\ & 2x + 10y + z = 13 \\ & 2x + 2y + 10z = 14 \end{aligned}$$

Solⁿ:

The coeff matrix of the given system of eqⁿs is in diagonally dominant form hence there is no need to rearrange the equations.

$$10x + y + z = 12$$

$$2x + 10y + z = 13$$

$$2x + 2y + 10z = 14$$

$$x = \frac{1}{10}(12 - y - z)$$

$$x = \frac{1}{10}(12 - y - z)$$

$$y = \frac{1}{10}(13 - 2x - z)$$

$$y = \frac{1}{10}(13 - 2x - z)$$

$$z = \frac{1}{10}(14 - 2x - 2y)$$

$$z = \frac{1}{10}(14 - 2x - 2y)$$

I approximation:

Let us assume that $(x, y, z) = (0, 0, 0)$

$$x' = \frac{1}{10}(12 - 0 - 0) = \frac{12}{10} = 1.2$$

$$y' = \frac{1}{10}(13 - 2(1.2) - 0) = \frac{13 - 2.4}{10} = 1.06$$

$$z' = \frac{1}{10}(14 - 2(1.2) - 2(1.06)) = \frac{14 - 2.4 - 2.12}{10} = 0.958$$

II approximation :

$$x'' = \frac{1}{10} (12 - (1.06) - (0.958)) = 0.9982$$

$$y'' = \frac{1}{10} (13 - 2(0.9982) - (0.958)) = 1.00456$$

$$z'' = \frac{1}{10} (14 - 2(0.9982) - 2(1.00456)) = 0.999448$$

$$x'' = 0.9982$$

$$y'' = 1.00456$$

$$z'' = 0.999448$$

III approximation :

$$x''' = \frac{1}{10} (12 - (1.00456) - (0.99945)) = 0.999599$$

$$y''' = \frac{1}{10} (13 - 2(0.999599) - (0.99945)) = 1.0001352$$

$$z''' = \frac{1}{10} (14 - 2(0.99959) - 2(1.00013)) = 1.000056$$

$$x''' = 0.9996$$

$$y''' = 1.0001$$

$$z''' = 1.0000$$

IV approximation:

$$x''' = \frac{1}{10} (12 - (1.0001) - (1.0000)) = \frac{9.999}{10} = 0.9999$$

$$y''' = \frac{1}{10} (13 - 2(0.9999) - (1.0000)) = 1.0000$$

$$z''' = \frac{1}{10} (14 - 2(0.9999) - 2(1.0000)) = 1.0000$$

V approximation:

$$x''' = 0.9999$$

$$y''' = 1.0000$$

$$z''' = 1.0000$$

$$x^{''''} = \frac{1}{10} (12 - 1 - 1) = 1.0000$$

$$y^{''''} = \frac{1}{10} (13 - 2(1) - 1) = 1.0000$$

$$z^{''''} = \frac{1}{10} (14 - 2(1) - 2(1)) = 1.0000$$

$$x^{''''} = 1.0000$$

$$y^{''''} = 1.0000$$

$$z^{''''} = 1.0000$$

Iterations	x	y	z
I	1.2	1.06	0.958
II	0.9982	1.00456	0.999448
III	0.999599	1.0001352	1.000056
IV	0.9999	1.0000	1.0000
V	1.0000	1.0000	1.0000

4. Solve the System of linear Equations.

(i) $x + 2y - z = 0$

$$2x + y + z = 0$$

$$x - 4y + 5z = 0$$

Solⁿ. Given System of Equations are homogeneous Eq^s

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 1 \\ 1 & -4 & 5 \end{bmatrix} \quad D = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$AX = D$$

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 2 & 1 & 1 & 0 \\ 1 & -4 & 5 & 0 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\boxed{n = 3}$$

$$\begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array} \quad \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 2 \\ 0 & -6 & 6 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 6R_2 \quad \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 2 \\ 0 & 0 & -6 \end{bmatrix}$$

$$\rho(A) = \boxed{3 = n}$$

as $n = r$

we got trivial solutions. $(0, 0, 0)$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

5) Test for consistency and solve them

(i) $x + y + 2z = 4$

(ii) $2x - y + 3z = 9$

(iii) $3x - y - z = 2$

Solⁿ Given system of eqⁿs are non homogeneous

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 3 \\ 3 & -1 & -1 \end{bmatrix} \quad D = \begin{bmatrix} 4 \\ 9 \\ 2 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$AD = X$$

$$\begin{bmatrix} 1 & 1 & 2 & 4 \\ 2 & -1 & 3 & 9 \\ 3 & -1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \boxed{n = 3}$$

$$\begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{array} \quad \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & -3 & -1 & 1 \\ 0 & -4 & -7 & -10 \end{bmatrix}$$

$$R_3 \rightarrow 3R_3 - 4R_2 \quad \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & -3 & -1 & 1 \\ 0 & 0 & -17 & -34 \end{bmatrix}$$

$$\rho(A) = \rho(AD)$$

$$\boxed{\lambda = 3}$$

$$\lambda = n ; \rho(A) = \rho(AD)$$

$\therefore$ Given system of eqⁿs are consistent.

$$-17z = -34 \Rightarrow z = 34/17 = 2$$

$$-3y - z = 1 \Rightarrow y = \frac{1+z}{-3} = -\frac{1}{3} \left[1 + \frac{34}{17} \right] = -1$$

$$x + y + 2z = 4 \Rightarrow x = 4 - 2z - y = 1$$

(4 - 4 + 1)

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

UNIT - 2

Q) Verify Cayley Hamilton theorem for

(i) $\begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$ Also find A^{-1} & A^4 .

Sol: The characteristic eqn of A is $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 0 & 3 \\ 2 & -1-\lambda & -1 \\ 1 & -1 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow -\lambda^3 + \lambda^2 + 5\lambda - 5 = 0$$

characteristic eqn of matrix A is $\lambda^3 - \lambda^2 - 5\lambda + 5 = 0$

→ To verify Cayley Hamilton theorem:-

$$\text{Let } \lambda = A \Rightarrow A^3 - A^2 - 5A + 5I = 0 \quad \text{--- (1)}$$

$$\begin{bmatrix} 4 & -3 & 21 \\ 9 & -8 & 1 \\ 5 & -5 & 5 \end{bmatrix} - \begin{bmatrix} 4 & -3 & 6 \\ -1 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix} + 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

∴ It satisfies Cayley Hamilton theorem

→ To find A^4

$$(A^3 - A^2 - 5A + 5I = 0) \times A$$

$$A^4 - A^3 - 5A^2 + 5A = 0$$

$$A^4 = A^3 + 5A^2 - 5A$$

$$A^4 = \begin{bmatrix} 4 & -3 & 21 \\ 9 & -8 & 1 \\ 5 & -5 & 5 \end{bmatrix} + 5 \begin{bmatrix} 4 & -3 & 6 \\ -1 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 19 & -18 & 36 \\ -6 & 7 & 36 \\ 0 & 0 & 25 \end{bmatrix}$$

→ To find A^{-1}

multiply A^{-1} with eq ①

$$A^2 - A - 5I + 5A^{-1} = 0$$

$$5A^{-1} = -A^2 + A + 5I \Rightarrow$$

$$A^{-1} = \frac{-A^2 + A + 5I}{5}$$

$$A^{-1} = - \begin{bmatrix} 4 & -3 & 6 \\ -1 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 3 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix} + 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 2/5 & 3/5 & -3/5 \\ 3/5 & 2/5 & -7/5 \\ 1/5 & -1/5 & 1/5 \end{bmatrix}$$

$$(ii) A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

Sol: The characteristic eqn of A is $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 2 & 3 \\ 2 & 4-\lambda & 5 \\ 3 & 5 & 6-\lambda \end{vmatrix} = 0$$

$$\Rightarrow -\lambda^3 + 11\lambda^2 + 4\lambda - 1 = 0$$

$$\Rightarrow \lambda^3 - 11\lambda^2 - 4\lambda + 1 = 0$$

To verify Cayley hamelton theorem:-

$$\text{Let } \lambda = A \rightarrow A^3 - 11A^2 - 4A + I = 0 \quad \text{--- (1)}$$

$$\begin{bmatrix} 157 & 283 & 353 \\ 283 & 570 & 636 \\ 353 & 636 & 793 \end{bmatrix} - 11 \begin{bmatrix} 14 & 25 & 31 \\ 25 & 45 & 56 \\ 31 & 56 & 70 \end{bmatrix} - 4 \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ It satisfies Cayley Hamelton theorem.

$\rightarrow$ To find A^4

$$\text{we have } (A^3 - 11A^2 - 4A + I = 0) \cdot \times A$$

$$A^4 - 11A^3 - 4A^2 + A = 0$$

$$A^4 = 11A^3 + 4A^2 - A$$

$$A^4 = 11 \begin{bmatrix} 157 & 283 & 353 \\ 283 & 570 & 636 \\ 353 & 636 & 793 \end{bmatrix} + 4 \begin{bmatrix} 14 & 25 & 31 \\ 25 & 45 & 56 \\ 31 & 56 & 72 \end{bmatrix} - \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1781 & 3211 & 4004 \\ 3211 & 5786 & 7216 \\ 4004 & 7216 & 8997 \end{bmatrix}$$

→ To find A^{-1}

multiply A^{-1} with eq (1)

$$A^2 - 11A - 4I + A^{-1} = 0$$

$$A^{-1} = -A^2 + 11A + 4I$$

$$A^{-1} = - \begin{bmatrix} 14 & 25 & 31 \\ 25 & 45 & 56 \\ 31 & 56 & 70 \end{bmatrix} + 11 \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} + 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 1 & -3 & 2 \\ -3 & 3 & -1 \\ 2 & -1 & 0 \end{bmatrix}$$

Q) Find modal matrix which transform the matrix $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$ into diagonal form & calculate A^4 .

Sol:- The characteristic eqn of A is $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{vmatrix} = 0$$

$$P^{-1}AP = D$$

$$\Rightarrow -\lambda^3 + 6\lambda^2 - 11\lambda + 6 = 0$$

$$\Rightarrow \lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

$$\lambda_1 = 1, \lambda_2 = 3, \lambda_3 = 2.$$

$\therefore$ Eigen values are 1, 2, 3

$\rightarrow$ Let $\lambda = 1$

$$|A - I|X = 0$$

$$\begin{bmatrix} 0 & 0 & -1 \\ 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$R_3 \leftrightarrow R_1 \begin{bmatrix} 2 & 2 & 2 \\ 1 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$R_2 \rightarrow 2R_2 - R_1 \begin{bmatrix} 2 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$R_2 \leftrightarrow R_3 \begin{bmatrix} 2 & 2 & 2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$r = 2, n = 3 \quad (r < n)$$

$$2x + 2y + 2z = 0$$

$$-z = 0 \Rightarrow z = 0$$

$$\text{Let } y = k \Rightarrow x = -k$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -k \\ k \\ 0 \end{bmatrix}$$

$\rightarrow$ Let $\lambda = 2$

$$|A - 2I|X = 0$$

$$\begin{bmatrix} -1 & 0 & -1 \\ 1 & 0 & 1 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$R_2 \rightarrow R_2 + R_1 \begin{bmatrix} -1 & 0 & -1 \\ 0 & 0 & 0 \\ 2 & 2 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 2R_1 \begin{bmatrix} -1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 2 & -1 \end{bmatrix}$$

$$R_2 \leftrightarrow R_3 \begin{bmatrix} -1 & 0 & -1 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$r = 2, n = 3 \quad (r < n)$$

$$-x - z = 0 \Rightarrow -x = z$$

$$2y - z = 0$$

$$\text{Let } y = k, z = 2k$$

$$x = -2k$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = k \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$$

let $\lambda = 3$

$$(A - 3I)x = 0$$

$$\begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\begin{aligned} R_2 &\rightarrow 2R_2 + R_1 \\ R_3 &\rightarrow R_3 + R_1 \end{aligned} \begin{bmatrix} -2 & 0 & -1 \\ 0 & -2 & 1 \\ 0 & 2 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2 \begin{bmatrix} -2 & 0 & -1 \\ 0 & -2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$r = 2 \quad n = 3 \quad (r < n)$$

$$-2x - z = 0 \Rightarrow x = -\frac{z}{2}$$

$$\text{Let } y = k$$

$$-2y + z = 0$$

$$z = 2k, \quad x = -k$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -k \\ k \\ 2k \end{bmatrix}$$

$$\text{modal matrix} = P = [x_1 \ x_2 \ x_3]$$

$$P = \begin{bmatrix} -1 & -2 & -1 \\ 1 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix}$$

$$P^{-1} = \begin{bmatrix} 0 & 1 & -1/2 \\ -1 & -1 & 0 \\ 1 & 1 & 1/2 \end{bmatrix}$$

$$\text{let } P^{-1}AP = D$$

$$\begin{aligned} P^{-1}AP &= \begin{bmatrix} 0 & 1 & -1/2 \\ -1 & -1 & 0 \\ 1 & 1 & 1/2 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix} \begin{bmatrix} -1 & -2 & -1 \\ 1 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \end{aligned}$$

$$PD P^{-1} = A$$

$$A^4 = PD^4 P^{-1}$$

$$= \begin{bmatrix} -1 & -2 & -1 \\ 1 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 81 \end{bmatrix} \begin{bmatrix} 0 & 1 & -1/2 \\ -1 & 7 & 0 \\ 1 & 1 & 1/2 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} -45 & -50 & -40 \\ 65 & 66 & 40 \\ 130 & 130 & 81 \end{bmatrix}$$

8) Reduce the Quadratic to canonical form.

Also find its rank, signature, nature, index

$$(1) 3x^2 + 5y^2 + 3z^2 - 2yz + 2zx - 2xy.$$

$$\text{Sol: } Q = X^T A X = \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 3-\lambda & -1 & 1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0 \Rightarrow -\lambda^3 + (\text{trace of } A)\lambda^2 - (\text{sum of co-factors of diagonal})\lambda + |A| = 0$$

$$\lambda_1 = 6, \lambda_2 = 3, \lambda_3 = 2$$

for $\lambda_1 = 6$

$$(A - 6I)X = 0$$

$$\begin{bmatrix} -3 & -1 & 1 \\ -1 & -1 & -1 \\ 1 & -1 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$R_2 \leftrightarrow R_3 \quad \begin{bmatrix} 1 & -1 & -3 \\ -1 & -1 & -1 \\ -3 & -1 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + R_1 \quad \begin{bmatrix} 1 & -1 & -3 \\ 0 & -2 & -4 \\ 0 & -4 & -8 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 3R_1$$

$$R_3 \rightarrow R_3 - 2R_2 \quad \begin{bmatrix} 1 & -1 & -3 \\ 0 & -2 & -4 \\ 0 & 0 & 0 \end{bmatrix}$$

$$r = 2 \quad n = 3$$

1 L I soln

$$\boxed{z = k}$$

$$-2y - 4z = 0 \Rightarrow \boxed{y = -2k}$$

$$\boxed{x = k}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = k \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = x_1$$

for $\lambda = 6$

$$\Rightarrow \text{for } \lambda_2 = 3$$

$$|A - 3I|X = 0$$

$$\begin{bmatrix} 0 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$R_3 \leftrightarrow R_1 \quad \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + R_1 \quad \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2 \quad \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$r = 2, \quad n = 3$$

$$\boxed{z = k}$$

$$y - z = 0$$

$$y = z$$

$$\boxed{y = k}$$

$$x - y = 0$$

$$\boxed{x = k}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = k \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = x_2 \quad \text{for } \lambda = 3$$

$$\Rightarrow P = [x_1 \ x_2 \ x_3]$$

$$P = \begin{bmatrix} 1 & 1 & -1 \\ -2 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\Rightarrow \lambda_3 = 2$$

$$|A - 2I|X = 0$$

$$\begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$R_2 \rightarrow R_2 + R_1 \quad \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_1$$

$$r = 2 \quad n = 3$$

$$1 \text{ L.I. soln}$$

$$\boxed{z = k}$$

$$2y = 0 \Rightarrow \boxed{y = 0}$$

$$x - y + z = 0$$

$$\boxed{x = -k}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} k = x_3 \quad \text{for } \lambda = 2$$

check

$$x_1 \cdot x_2 = 1 - 2 + 1 = 0$$

$$x_2 \cdot x_3 = -1 + 0 + 1 = 0$$

$$x_3 \cdot x_1 = -1 + 0 + 1 = 0$$

$$\therefore P^{-1} = P^T$$

→ finding normalized matrix "P"

$$P = \begin{bmatrix} \frac{x_1}{\|x_1\|} & \frac{x_2}{\|x_2\|} & \frac{x_3}{\|x_3\|} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} \\ -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\rightarrow P^T A P = \begin{bmatrix} \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} \\ -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix} = D \quad \therefore \underline{P^T A P = D}$$

$$Y^T D Y = 6x^2 + 3y^2 + 2z^2 = 0$$

$$\therefore \text{index} = 3 (6, 3, 2)$$

$$\text{Rank} = 3$$

$$\begin{aligned} \text{Signature} &= 2S - r \\ &= 2(3) - 3 = 3 \end{aligned}$$

nature = +ve definite (all 3 are +ve)

Q) Reduce quadratic to Canonical form

(ii) $2x^2 + 2y^2 + 2z^2 - 2xy - 23x - 24z$

Sol:- $Q^T A Q$ $Q = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ $Q^T = [x \ y \ z]$

$$A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

characteristic eq $|A - \lambda I| = 0$

$$\begin{vmatrix} 2-\lambda & -1 & -1 \\ -1 & 2-\lambda & -1 \\ -1 & -1 & 2-\lambda \end{vmatrix} = 0$$

$$-\lambda^3 + 6\lambda^2 - (3+3+3)\lambda + 0 = 0$$

$$\lambda^3 - 6\lambda^2 + 9\lambda = 0$$

$$\lambda_1 = 3, \lambda_2 = 3, \lambda_3 = 0$$

→ Eigen values are 3, 3, 0

→ Eigen vectors for $\lambda = 3$

Let $(A - 3I)X = 0$

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & -1 & -1 \\ -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$R_2 \rightarrow R_2 - R_1 \quad R_3 \rightarrow R_3 - R_1 \quad \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$r=1 \quad n=3 \quad n-r=2 \quad L.I$$

$$-x - y - z = 0$$

$$y = k_1, \quad z = k_2, \quad x = -k_1 - k_2$$

$$X = \begin{bmatrix} -k_1 \\ k_1 \\ 0 \end{bmatrix} + \begin{bmatrix} -k_2 \\ 0 \\ k_2 \end{bmatrix}$$

$$X_1 = k_1 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \quad X_2 = k_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

→ Eigen vectors for $\lambda = 0$

$$(A - 0I)X = 0$$

$$\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$R_2 \rightarrow 2R_2 + R_1 \quad R_3 \rightarrow 2R_3 + R_1 \quad \begin{bmatrix} 2 & -1 & -1 \\ 0 & 3 & -3 \\ 0 & -3 & 3 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2 \quad \begin{bmatrix} 2 & -1 & -1 \\ 0 & 3 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$n=3, \quad r=2 \quad n-r=1$$

$$2x - y - z = 0$$

$$3y - 3z = 0$$

$$\Rightarrow z = k, \quad y = k \quad X_3 = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = k \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$2x - y - z = 0 \Rightarrow x = k$$

$$P = [x_1 \ x_2 \ x_3]$$

$$P = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$x_1 x_2 = 1, \quad x_2 x_3 = -1+1 = 0$$

$$x_3 x_1 = -1+1 = 0$$

$$P = \begin{bmatrix} a & -1 & 1 \\ b & 0 & 1 \\ c & 1 & 1 \end{bmatrix}$$

$$-a+c=0$$

$$a+b+c=0$$

$$c=a$$

$$b=-(a+c)$$

$$c=k$$

$$b=-2k$$

$$a=k$$

$$x_1 = k \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$$x_1 x_2 = -1+1=0$$

$$x_2 x_3 = -1+1=0$$

$$x_3 x_1 = -2+1+1=0$$

$$P = \begin{bmatrix} 1 & -1 & 1 \\ -2 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

P is orthogonal matrix

$$\therefore \boxed{P^{-1} = P^T}$$

normalization of matrix

$$P = \begin{bmatrix} \frac{x_1}{\|x_1\|} & \frac{x_2}{\|x_2\|} & \frac{x_3}{\|x_3\|} \end{bmatrix}$$

$$P = \begin{bmatrix} 1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3} \\ -2/\sqrt{6} & 0 & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \end{bmatrix}$$

$$P^T A P = \begin{bmatrix} 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3} \\ -2/\sqrt{6} & 0 & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix} = D$$

$$\therefore \underline{P^T A P = D}$$

②

$$Y^T D Y = 3x^2 + 3y^2$$

$$\text{index} = 2$$

$$\text{Rank} = 2$$

$$\begin{aligned}\text{Signature} &= 2(2) - 2 \\ &= 2\end{aligned}$$

$$\text{nature} = +ve \text{ Semidefinite}$$

UNIT-3

1Q. State Rolle's theorem and verify $f(x) = (x+2)^3(x-3)^4$ in the interval $[-2, 3]$

Statement:- If a function $f(x)$

1. is continuous on $[a, b]$
2. differentiable on (a, b)
3. $f(a) = f(b)$

Then there exists $c \in (a, b)$ such that $f'(c) = 0$.

Verification for $f(x) = (x+2)^3(x-3)^4$

$f(x)$ is a polynomial $\Rightarrow$ its continuous and differentiable everywhere

$$f(-2) = 0 \text{ and } f(3) = 0$$

$$f(-2) = f(3)$$

as it verifies Rolle's conditions

$$f'(c) = 0$$

$$f'(c) = 3(c+2)^2(c-3)^4 + 4(c+2)^3(c-3)^3$$

$$f'(c) = (c+2)^2(c-3)^3(7c-1)$$

$$f'(c) = 0$$

$$c = -2, 3, 1/7$$

as the c value lies in $(-2, 3)$

$$\therefore \boxed{c = 1/7}$$

Q. (i) If $a < b$ P.T $\frac{b-a}{1+b^2} < \tan^{-1}b - \tan^{-1}a < \frac{b-a}{1+a^2}$

let $f(x) = \frac{1}{1+x^2}$

According to Lagrange's, there exists $c \in (a, b)$.

Such that $\tan^{-1}b - \tan^{-1}a = \frac{b-a}{1+c^2}$

According to inequality.

$$a < c < b$$

$$s.o.b.s$$

$$b^2 > c^2 > a^2$$

adding 1 on each side

$$1+b^2 > 1+c^2 > 1+a^2$$

taking reciprocal

$$\frac{1}{1+b^2} < \frac{1}{1+c^2} < \frac{1}{1+a^2}$$

multiply by $(b-a)$

$$\frac{b-a}{1+b^2} < \tan^{-1}b - \tan^{-1}a < \frac{b-a}{1+a^2} //$$

(ii) $f(x) = \tan^{-1}(4/3)$

let $a=1, b=4/3$

according to the above inequality.

$$\frac{b-a}{1+b^2} < \tan^{-1}b - \tan^{-1}a < \frac{b-a}{1+a^2}$$

$$b-a = 1/3, 1+b^2 = 1+16/9 = 25/9.$$

$$\frac{1/3}{25/9} < \tan^{-1}(4/3) < \frac{1/3}{2}$$

Since $\tan^{-1}(1) = \pi/4$.

$$\pi/4 + 3/25 < \tan^{-1}(4/3) < \pi/4 + 1/6$$

3Q. Verify Lagrange's mean value theorem for the function $f(x)$

Statement: If a function $f(x)$ satisfies:

1. $f(x)$ is continuous in $[a, b]$.
2. $f(x)$ is differentiable in (a, b) .

then,

$$f'(c) = \frac{f(b) - f(a)}{b - a} \quad \left| \begin{array}{l} \text{for } f(x) = \cos x \\ f'(x) = -\sin x \\ \frac{f(b) - f(a)}{b - a} = \frac{0 - 1}{\pi/2} = -2/\pi \\ f'(c) = -2/\pi \Rightarrow \sin c = 2/\pi \end{array} \right.$$

4Q. Verify Cauchy's for $f(x) = \sqrt{x}$, $g(x) = 1/\sqrt{x}$ in the interval $[a, b]$.

According to Cauchy's

1. $f(x)$ and $g(x)$ are continuous on $[a, b]$
2. $f(x)$ and $g(x)$ are differentiable on (a, b)

$$f'(x) = 1/2\sqrt{x}, \quad g'(x) = -\frac{1}{2x\sqrt{x}}$$

according to CMVT

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$$

$$\frac{\sqrt{b} - \sqrt{a}}{1/\sqrt{b} - 1/\sqrt{a}} = -\frac{1/2\sqrt{c}}{1/2c\sqrt{c}}$$

$$\sqrt{ab} = c$$

$$c = \sqrt{ab}$$

5) We state Taylors series and expand $f(x) = \log(\sin x)$ in powers $x-2$.

If $f(x)$ has derivative of all orders

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \dots$$

expansion of $f(x) = \log(\sin x)$ about $x=2$

$$f(x) = \log(\sin x)$$

Derivatives

$$f'(x) = \cot x.$$

$$f''(x) = -\operatorname{cosec}^2 x.$$

values at $x=2$

$$f(2) = \log(\sin 2),$$

$$f'(2) = \cot 2$$

$$f''(2) = -\operatorname{cosec}^2 2$$

Taylors expansion in powers of $(x-2)$

$$\log(\sin x) = \log(\sin 2) + \cot 2 (x-2) - \frac{\operatorname{cosec}^2 2}{2} (x-2)^2$$

$\therefore$ expansion about $x=2$ is obtained.

6Q. state mclaren series for $f(x) = \log(1+x)$

if $f(x)$ has derivatives of all orders at $x=0$, then

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \dots$$

mclaurin for $f(x) = \log(1+x)$

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

UNIT-4

①

① If $x^x y^y z^z = e$ then show that $\frac{\partial^2 z}{\partial x \partial y} = -(\log e)^{-1}$

if $x=y=z$.

$$x^x y^y z^z = e$$

— Apply log on both sides

$$\log(x^x y^y z^z) = \log e$$

$$\log x^x + \log y^y + \log z^z = 1$$

$$x \log x + y \log y + z \log z = 1$$

$$\frac{\partial}{\partial x} (x \log x) + \frac{\partial}{\partial x} (y \log y) + \frac{\partial}{\partial x} (z \log z) = (1) \frac{\partial}{\partial x}$$

$$\log x + x \cdot \frac{1}{x} + 0 + \frac{\partial z}{\partial x} \log z + z \cdot \frac{1}{z} \frac{\partial z}{\partial x} = 0$$

$$(\log x + 1) + \frac{\partial z}{\partial x} (\log z + 1) = 0$$

$$\frac{\partial z}{\partial x} = \frac{-(\log x + 1)}{\log z + 1}$$

$$\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = -(\log x + 1) \cdot (-1) (1 + \log z)^{-2} \cdot \frac{1}{z} \frac{\partial z}{\partial y}$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\log x + 1}{z (1 + \log z)^2} \frac{\partial z}{\partial y} \rightarrow \text{①}$$

$$\frac{\partial^2}{\partial y} (x \log x) + \frac{\partial}{\partial y} (y \log y) + \frac{\partial}{\partial y} (z \log z) = \frac{\partial}{\partial y} \quad (1)$$

$$0 + y \frac{1}{y} + \log y + \frac{\partial z}{\partial y} \log z + \frac{1}{z} z \frac{\partial z}{\partial y} = 0$$

$$(\log y + 1) + \frac{\partial z}{\partial y} (\log z + 1) = 0$$

$$\frac{\partial z}{\partial y} = \frac{-(\log y + 1)}{\log z + 1} \rightarrow (2)$$

Sub (2) in (1)

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{(\log x + 1)}{z(\log z + 1)^2} - \frac{(\log y + 1)}{(\log z + 1)}$$

Given $x = y = z$ from

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{-1}{x(\log x + 1)}$$

$$= -[x(\log x + \log e)]^{-1}$$

$$\frac{\partial^2 z}{\partial x \partial y} = -[x \log e x]^{-1}$$

∴ Hence proved

② If $U = \frac{x+y}{1-xy}$ $V = \tan^{-1}x + \tan^{-1}y$ then find Jacobian UV w.r.to x & y and relation b/w them.

$$J\left(\frac{U, V}{x, y}\right) = \begin{vmatrix} \frac{\partial U}{\partial x} & \frac{\partial U}{\partial y} \\ \frac{\partial V}{\partial x} & \frac{\partial V}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{(1-xy)(1) - (x+y)(-y)}{(1-xy)^2} & \frac{(1-xy)(1) + (x+y)(-x)}{(1-xy)^2} \\ \frac{1}{1+x^2} & \frac{1}{1+y^2} \end{vmatrix}$$

$$= \frac{(1+y^2)}{(1-xy)^2} \cdot \frac{1}{(1+y^2)} - \frac{(1+x^2)}{(1-xy)^2} \cdot \frac{1}{1+x^2} = 0$$

$J\left(\frac{UV}{xy}\right) = 0$ then U, V are functionally dependent

Functionally dependent :-

Let $u = u(x, y)$ $v = v(x, y)$ be two functions if Jacobian of $J\left(\frac{UV}{xy}\right) = 0$ then u, v are said to be functionally dependent in this case one function can be defined in terms of another function.

$$V = \tan^{-1}x + \tan^{-1}y$$

$$V = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

$$V = \tan^{-1}(U) \text{ — the relation b/w } U \text{ \& } V$$

8) Rectangular box open at the top is to have a volume 32 cubic units find the dimensions of the box required for its construction

$$f = lb + 2bh + 2lh \quad \phi = lbh - 32$$

$\rightarrow (i) \qquad \qquad \qquad \rightarrow (ii)$

$$-F = f + \lambda \phi = lb + 2bh + 2lh + \lambda(lbh - 32) \rightarrow (iii)$$

$$\frac{\partial F}{\partial l} = b + 2h + \lambda(bh) = 0 \quad \frac{b+2h}{bh} = -\lambda \rightarrow (1)$$

$$\frac{\partial F}{\partial b} = l + 2h + \lambda(lh) = 0 \quad \frac{2h+l}{lh} = -\lambda \rightarrow (2)$$

$$\frac{\partial F}{\partial h} = 2b + 2l + \lambda(lb) = 0 \quad \frac{2b+2l}{lb} = -\lambda \rightarrow (3)$$

$$(1) = (2)$$

$$\frac{b+2h}{bh} = \frac{2h+l}{lh}$$

$$lb + 2hl = 2hb + lb$$

$$2hl = 2hb$$

$$l = b$$

$$(2) = (3)$$

$$\frac{2h+l}{lh} = \frac{2b+2l}{lb}$$

$$2hb + lb = 2bh + 2lb$$

$$lb = 2lb$$

$$h = b/2$$

$$\text{As } lbh = 32$$

$$b \times b \times b/2 = 32$$

$$b^3 = 64 \quad b = 4 \rightarrow (4)$$

$$l = 4 \quad b = 4 \quad h = 2$$

sub in eqn (i)

$$-f = 16 + 16 + 16 = 48$$

- ④ Divide 40 into three +ve parts such that product of the first cube of second, square of third.

$$F = x + y + z + \lambda$$

$$\phi = x + y + z - 40 \quad f = xy^3z^2$$

$$F = xy^3z^2 + \lambda(x + y + z - 40) \rightarrow (1)$$

$$\frac{\partial F}{\partial x} = y^3z^2 + \lambda = 0$$

$$-\lambda = y^3z^2 \rightarrow (2)$$

$$\frac{\partial F}{\partial y} = 3xy^2z^2 + \lambda = 0$$

$$-\lambda = 3xy^2z^2 \rightarrow (3)$$

$$\frac{\partial F}{\partial z} = 2xy^3z + \lambda = 0$$

$$-\lambda = 2xy^3z \rightarrow (4)$$

$$(2) = (3)$$

$$y^3z^2 = 3xy^2z^2$$

$$y = 3x$$

$$x = y/3 \rightarrow (i)$$

$$(3) = (4)$$

$$3xy^2z^2 = 2xy^3z$$

$$3z = 2y$$

$$z = \frac{2y}{3} \rightarrow (ii)$$

sub eq (i) & (ii) in eq $x + y + z = 40$

$$y/3 + y + \frac{2y}{3} = 40$$

$$y + 3y + 2y = 120$$

$$6y = 120$$

$$y = 20 \quad x = \frac{20}{3} \quad z = \frac{40}{3}$$

$$xy^3z^2 = \frac{20}{3} \frac{(20)^3 (40)^2}{3^2} = \frac{(40)^2 (20)^4}{3^3}$$

⑤

$\int_0^{\pi/2} \sin^5 \theta \cos^{1/2} \theta d\theta$ in terms of β function

$$\int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta = \frac{1}{2} \beta(m, n)$$

$$2m-1=5 \quad 2n-1=1/2$$

$$m=3$$

$$n=9/4$$

$$\int_0^{\pi/2} \sin^{2(3)-1} \theta \cos^{2(9/4)-1} \theta d\theta = \frac{1}{2} \beta(3, 9/4)$$

$$\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

$$\frac{1}{2} \beta(3, 9/4) = \frac{1}{2} \frac{\Gamma(3) \Gamma(9/4)}{\Gamma(21/4)} = \frac{1}{2} \frac{2! \Gamma(9/4)}{\Gamma(21/4)} = 2)$$

$$= \frac{\Gamma(9/4)}{\frac{17}{4} \frac{13}{4} \frac{9}{4} \Gamma(9/4)} = \frac{4^3}{17 \times 13 \times 9} = \frac{64}{1989}$$

⑥

$$\int_0^{\infty} x^4 e^{-x^2} dx$$

$$\int_0^{\infty} x^{n-1} e^{-x} dx = \Gamma(n)$$

$$\text{let } x^2 = t \quad x = t^{1/2}$$

$$2x dx = dt$$

$$x dx = \frac{dt}{2}$$

x	0	∞
t	0	∞

$$= \int_0^{\infty} x^3 e^{-x^2} x dx$$

$$= \int_0^{\infty} t^{3/2} e^{-t} \frac{dt}{2}$$

$$= \frac{1}{2} \int_0^{\infty} t^{3/2} e^{-t} dt \Rightarrow \frac{1}{2} \int_0^{\infty} t^{5/2-1} e^{-t} dt = \frac{1}{2} \Gamma(5/2)$$

$$= \frac{1}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma_{1/2} = \frac{3}{8} \sqrt{\pi}$$

⑦ P.T $\int_0^1 x^{n-1} [\log(1/x)]^{m-1} dx = \frac{\Gamma_m}{n^m}$

$$\Gamma_m = \int_0^{\infty} e^{-x} x^{n-1} dx$$

let $\log 1/x = t \Rightarrow 1/x = e^t$

$$x = e^{-t}$$

$$dx = -e^{-t} dt$$

x	0	1
t	∞	0

$$= \int_{\infty}^0 (e^{-t})^{n-1} (t)^{m-1} - (e^{-t}) dt$$

$$= \int_0^{\infty} e^{-tn+t} (t)^{m-1} dt$$

$$= \int_0^{\infty} e^{-tn} t^{m-1} dt$$

let $\Rightarrow -tn = y$

$$-t = y/n \Rightarrow dt = dy/n$$

$$= \int_0^{\infty} e^{-y} (y/n)^{m-1} \frac{dy}{n}$$

$$= \frac{1}{n^{m-1+1}} \int_0^{\infty} e^{-y} (y)^{m-1} dy$$

$$= \frac{1}{n^m} \Gamma_m \Rightarrow \frac{\Gamma_m}{n^m}$$

$\therefore$ — Hence proved

① Evaluate $\int_0^3 \int_0^{x^2} x[x^2+y^2] dx dy$

$$\text{sol: } = \int_0^3 x^3 (y)_0^{x^2} + \left(\frac{xy^2}{3}\right)_0^{x^2} dx$$

$$= \int_0^3 x^3 x^2 + \frac{xx^6}{3} dx$$

$$= \int_0^3 x^5 + \frac{x^7}{3} dx$$

$$= \int_0^3 \left[\frac{x^6}{6} + \frac{x^8}{24} \right]$$

$$= \left(\frac{x^6}{6} \right)_0^3 + \left(\frac{x^8}{24} \right)_0^3$$

$$= \frac{3^6 \times 4 + 3^8}{24}$$

$$= \frac{3^6(13)}{24} //$$

② change of order $\int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} dy dx$

sol:- Given:-

$$x = 0 \text{ to } 4a$$

$$y = x^2/4a \text{ to } 2\sqrt{ax}$$

change

$$x = y^2/4a \text{ to } 2\sqrt{ay}$$

$$y = 0 \text{ to } 4a$$

$$= \int_0^{4a} \int_{y^2/4a}^{2\sqrt{ay}} dx dy$$

$$= \int_0^{4a} 2\sqrt{ay} \left[x \right]_{y^2/4a}^{2\sqrt{ay}} dy$$

$$= \int_0^{4a} 2\sqrt{ay} \cdot \frac{y^2}{4a} dy$$

$$= \left[\frac{2\sqrt{a} \cdot (y^3/2)}{3/2} - \frac{y^3}{4a} \right]_0^{4a}$$

$$= \left[\frac{4\sqrt{a} \cdot y^{3/2}}{3} - \frac{y^3}{12a} \right]_0^{4a}$$

$$= \frac{a \cdot 16\sqrt{a} \sqrt{y} - y^3}{12a}$$

$$= \frac{16a\sqrt{a} \cdot 4a \cdot \sqrt{4a} - 4^3 a^3}{12a}$$

$$= \frac{(16 \times 8)a^3 - 4a \cdot \sqrt{4a} - 4^3 a^3}{12a}$$

$$= \frac{128a^3 - 64a^3}{12a}$$

$$= \frac{\cancel{32}^{16} \cancel{4}^2}{\cancel{12}^3}$$

$$= \frac{16a^2}{3} //$$

$$\textcircled{3} \int_0^1 \int_{x^2}^x xy(x+y) dy dx$$

$$\text{Sol:} \int_0^1 \int_{x^2}^x x[x+y] dx dy$$

$$= \int_0^1 \left[\int_{x^2}^x x^2 y + xy^2 dy \right] dx$$

$$= \int_0^1 \left[\frac{x^2 (y)^2}{2} + \frac{xy^3}{3} \right]_{x^2}^x dx$$

$$= \int_0^1 \frac{x^2 x^2}{2} + \frac{xx^3}{3} - \frac{x^2 x^4}{2} - \frac{xx^6}{3} dx$$

$$= \int_0^1 \frac{5x^4}{6} - \frac{x^6}{2} - \frac{x^7}{3} dx$$

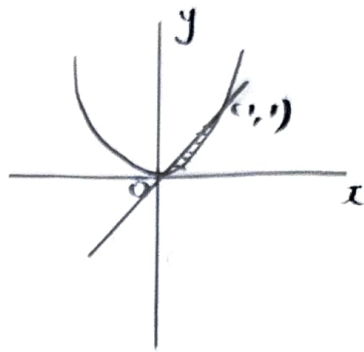
$$= \frac{x^5}{5} \Big|_0^1 - \frac{(x^7)}{7 \times 2} \Big|_0^1 - \frac{(x^8)}{8 \times 3} \Big|_0^1$$

$$= \frac{4}{24} - \frac{1}{14} - \frac{1}{24}$$

$$= \frac{3}{24} - \frac{1}{14}$$

$$= \frac{18}{306}$$

$$= 3/56 //$$



④ Evaluate $\iiint_V dx dy dz$ in finite region of space bounded

by $x=0, y=0, z=0$ & $2x+3y+4z=12$.

Sol:-

$$z = 0 \text{ to } \frac{12-2x-3y}{4}$$

$$y = 0 \text{ to } \frac{12-2x}{3}$$

$$x = 0 \text{ to } 6$$

$$= \int_0^6 \int_0^{\frac{12-2x}{3}} \int_0^{\frac{12-2x-3y}{4}} dx dy dz$$

$$= \int_0^6 \int_0^{\frac{12-2x}{3}} \int_0^{\frac{12-2x-3y}{4}} \frac{1}{4} [12-3y-2x] dy dx$$

$$= \frac{1}{4} \int_0^6 \left[\frac{1}{2} (12-2x)(y)_0^{\frac{12-2x}{3}} - \frac{3}{2} (y^2)_0^{\frac{12-2x}{3}} \right] dx$$

$$= \frac{1}{4} \int_0^6 (12-2x)^2 \left(\frac{1}{3} \right) - \frac{1}{8} (12-2x)^2 dx$$

$$= \frac{1}{4} \times \frac{1}{8} \int_0^6 (12-2x)^2 dx \longrightarrow \text{taking } 12-2x=t$$

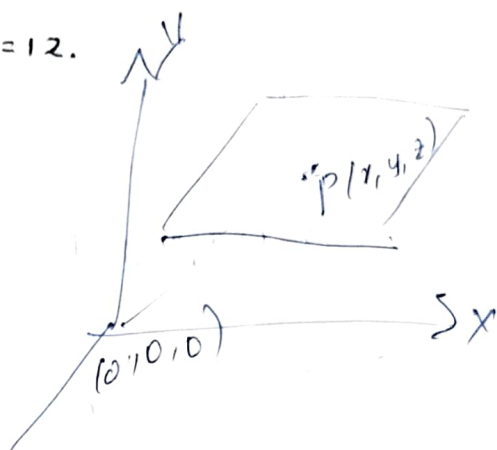
$$= \frac{1}{4} \times \frac{1}{6 \times 2} \int_{12}^0 t^2 dt$$

$$0-2dx=dt$$

$$= \frac{1}{24 \times 2} \left(\frac{t^3}{3} \right)_0^{12}$$

$$= \frac{(12)^3}{12 \times 4 \times 3}$$

$$= 12 //$$



x	0	6
t	12	0

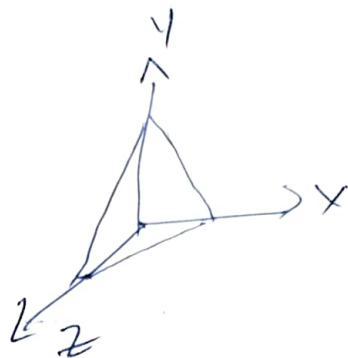
5) Find the volume of the tetrahedron on the coordinate planes.

$$\iiint dx dy dz \quad \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

$$z = 0 \text{ to } c - \frac{c}{a}x - \frac{c}{b}y$$

$$y = 0 \text{ to } b - \frac{b}{a}x$$

$$x = 0 \text{ to } a$$



$$= \int_0^a \int_0^{b-\frac{b}{a}x} \int_0^{c-\frac{c}{a}x-\frac{c}{b}y} dz dy dx$$

$$= \int_0^a \int_0^{b-\frac{b}{a}x} \left(c - \frac{c}{a}x - \frac{c}{b}y \right) dy dx$$

$$= \int_0^a c \left[y - \frac{x}{a}y - \frac{y^2}{2b} \right]_0^{b-\frac{b}{a}x} dx$$

$$= \int_0^a c \left(b - \frac{b}{a}x \right) \left(1 - \frac{x}{a} \right) - \frac{b^3}{2b} \left(1 - \frac{x}{a} \right)^2 dx$$

$$= \int_0^a \frac{cb}{2} \left(1 - \frac{x}{a} \right)^2 dx$$

$$\Rightarrow \text{taking } \left(1 - \frac{x}{a} \right) = t$$

$$\left(0 - \frac{dx}{a} \right) = dt$$

$$-\frac{dx}{a} = dt$$

$$\Rightarrow -\frac{a^3 x}{2} \int_1^0 t^2 du = \frac{abc}{2} \int_0^1 t^2 du$$

$$\Rightarrow \frac{abc}{2} \left(\frac{t^3}{3} \right)_0^1 = \frac{abc}{6} //$$

⑥ Evaluate the $\iiint dx dy dz$ over the sphere $x^2 + y^2 + z^2 = a^2$

$$\begin{aligned} \text{Sol: } z &= 0 \text{ to } \sqrt{a^2 - x^2 - y^2} \\ y &= 0 \text{ to } \sqrt{a^2 - x^2} \\ x &= 0 \text{ to } a \end{aligned} \quad \Rightarrow \int_0^a \int_0^{\sqrt{a^2 - x^2}} \int_0^{\sqrt{a^2 - x^2 - y^2}} dz dy dx$$

$$\Rightarrow \int_0^a \int_0^{\sqrt{a^2 - x^2}} \sqrt{a^2 - x^2 - y^2} dy dx$$

$$\Rightarrow \int_0^{\pi/2} \int_0^a \sqrt{a^2 - r^2} r dr d\theta$$

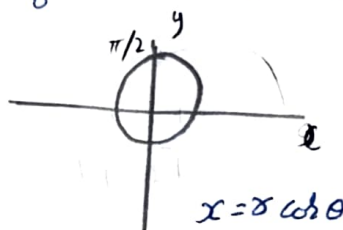
$$\Rightarrow \int_0^{\pi/2} \int_{a^2}^0 \sqrt{u} \left(-\frac{1}{2}\right) du d\theta$$

$$\Rightarrow \frac{1}{2} \int_0^{\pi/2} \int_0^{a^2} \sqrt{u} du d\theta$$

$$\Rightarrow \frac{1}{2} \times \frac{2}{3} \int_0^{\pi/2} (u^{3/2})_0^{a^2} d\theta$$

$$\Rightarrow \frac{1}{3} \int_0^{\pi/2} a^3 d\theta$$

$$\Rightarrow a^3 \left(\frac{\pi}{2}\right) \times \frac{1}{3} = \frac{a^3 \pi}{6} \Rightarrow \frac{\pi a^3}{6}$$



$$y = r \sin \theta$$

$$r^2 = x^2 + y^2$$

$$x^2 + y^2 = a^2$$

$$a^2 - r^2 = u$$

$$du = -2r dr$$

r	0	a
u	a^2	0